

F. Substrings in a String

time limit per test: 6 seconds
memory limit per test: 256 megabytes

Given a string s , process q queries, each having one of the following forms:

- 1 $i\ c$ — Change the i -th character in the string to c .
- 2 $l\ r\ y$ — Consider the substring of s starting at position l and ending at position r . Output the number of times y occurs as a substring in it.

Input

The first line of the input contains the string s ($1 \leq |s| \leq 10^5$) of lowercase English letters.

The second line contains an integer q ($1 \leq q \leq 10^5$) — the number of queries to process.

The next q lines describe the queries and may have one of the following forms:

- 1 $i\ c$ ($1 \leq i \leq |s|$)
- 2 $l\ r\ y$ ($1 \leq l \leq r \leq |s|$)

c is a lowercase English letter and y is a non-empty string consisting of only lowercase English letters.

The sum of $|y|$ over all queries of second type is at most 10^5 .

It is guaranteed that there is at least one query of second type.

All strings are 1-indexed.

$|s|$ is the length of the string s .

Output

For each query of type 2, output the required answer in a separate line.

Examples

input	Copy
ababababa 3 2 1 7 aba 1 5 c 2 1 7 aba	
output	Copy
3 1	

input	Copy
abdcdbc 5 2 1 7 bc 1 4 b 2 4 7 bc 1 2 a 2 1 4 aa	
output	Copy
2 2 1	

Note

Consider the first sample case. Initially, the string `aba` occurs 3 times in the range $[1, 7]$. Note that two occurrences may overlap.

After the update, the string becomes `ababcbaba` and now `aba` occurs only once in the range $[1, 7]$.

Codecraft-18 and Codeforces Round 458 (Div. 1 + Div. 2, combined)

Finished

Practice

Virtual participation

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Start virtual contest

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You can clone this contest to a mashup.

Clone Contest

Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

Last submissions

Submission	Time	Verdict
365169903	Mar/03/2026 08:03	Accepted
365169551	Mar/03/2026 07:58	Accepted

Problem tags

bitmasksbruteforcedata structuresstring suffix structuresstrings*3000

No tag edit access

Contest materials

Announcement (en)

Tutorial (en)